

IMAGE SCALING

Computer Vision with Hüseyin Özdemir

Linear Transformation equation for scaling (resizing)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Linear Transformation equation for scaling (resizing)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Transformed coordinate vector

Scale matrix

Input coordinate vector

Linear Transformation equation for scaling (resizing)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$x' = \alpha x$$

$$y' = \beta y$$

Linear Transformation equation for scaling (resizing)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$x' = \alpha x$$

$$y' = \beta y$$

α and β are scale parameters for horizontal and vertical directions respectively

Linear Transformation equation for scaling (resizing)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

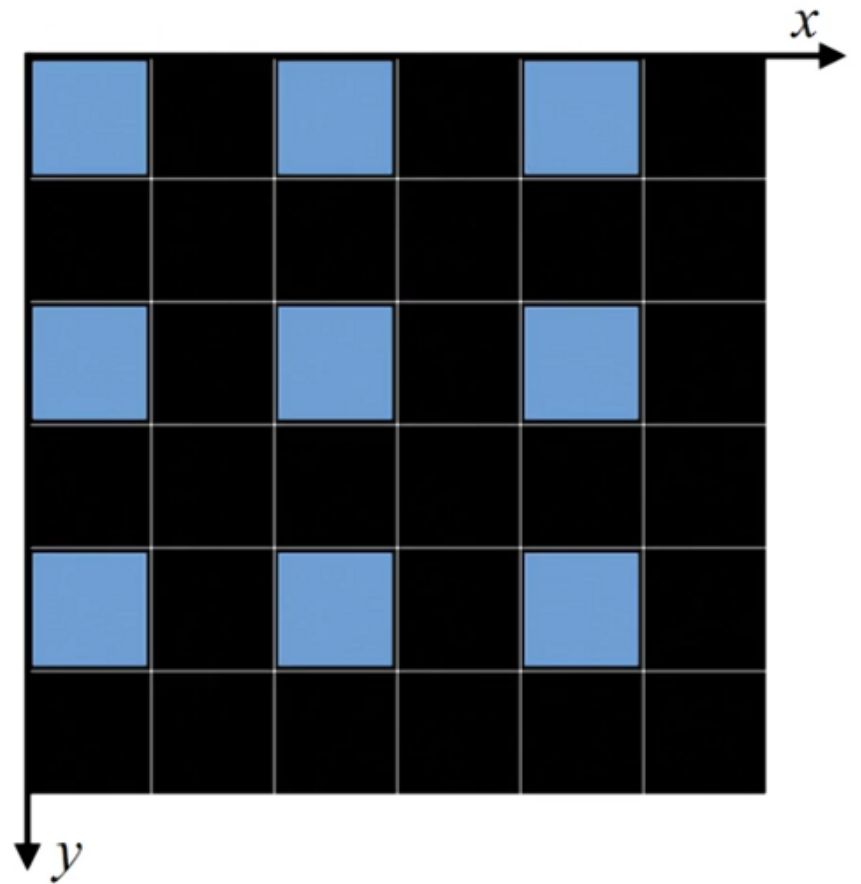
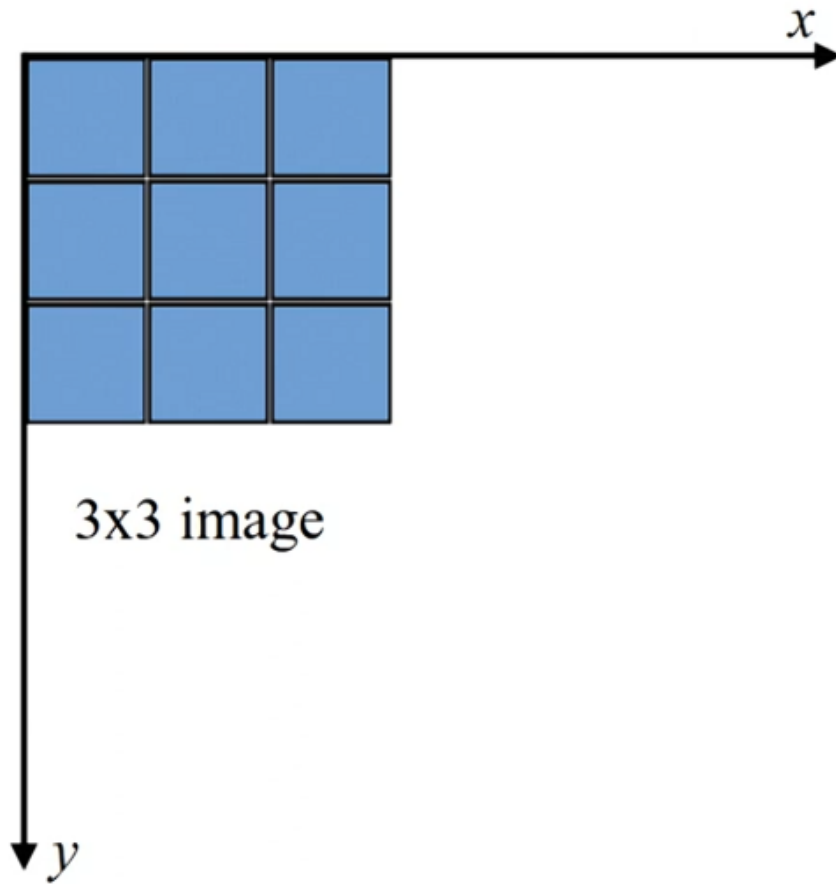
$$x' = \alpha x$$

$$y' = \beta y$$

If α is equal to β , then transformation is
isotropic (uniform) scale

This is the equation to scale in
both directions by 2

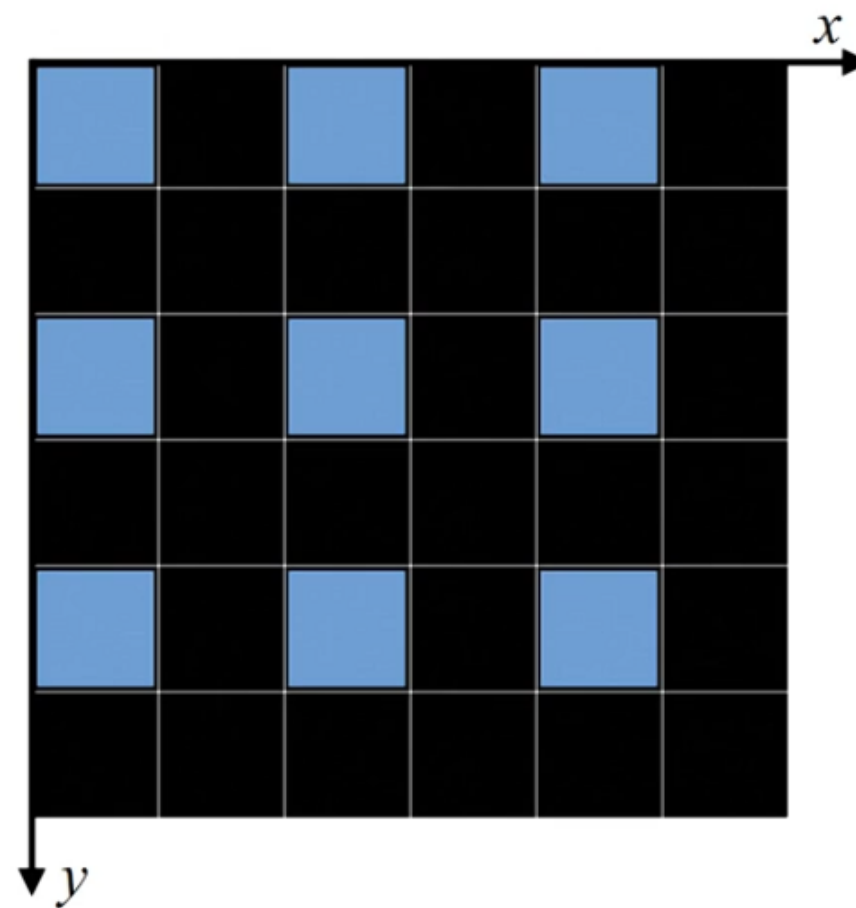
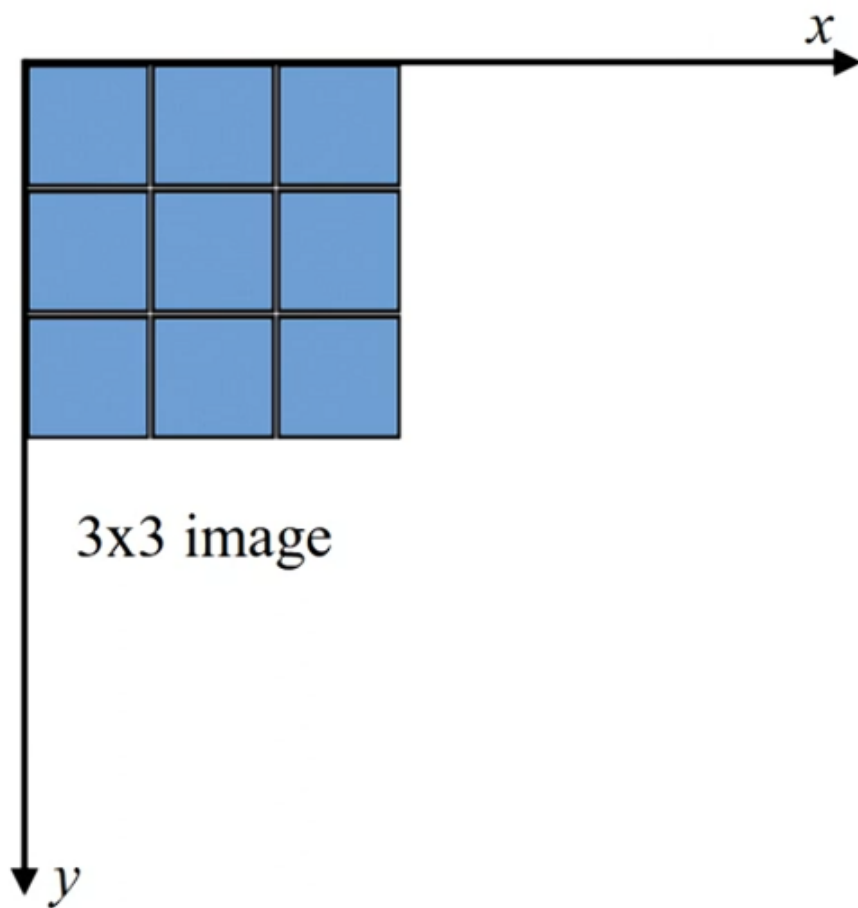
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



If input image is transformed directly

This is the equation to scale in
both directions by 2

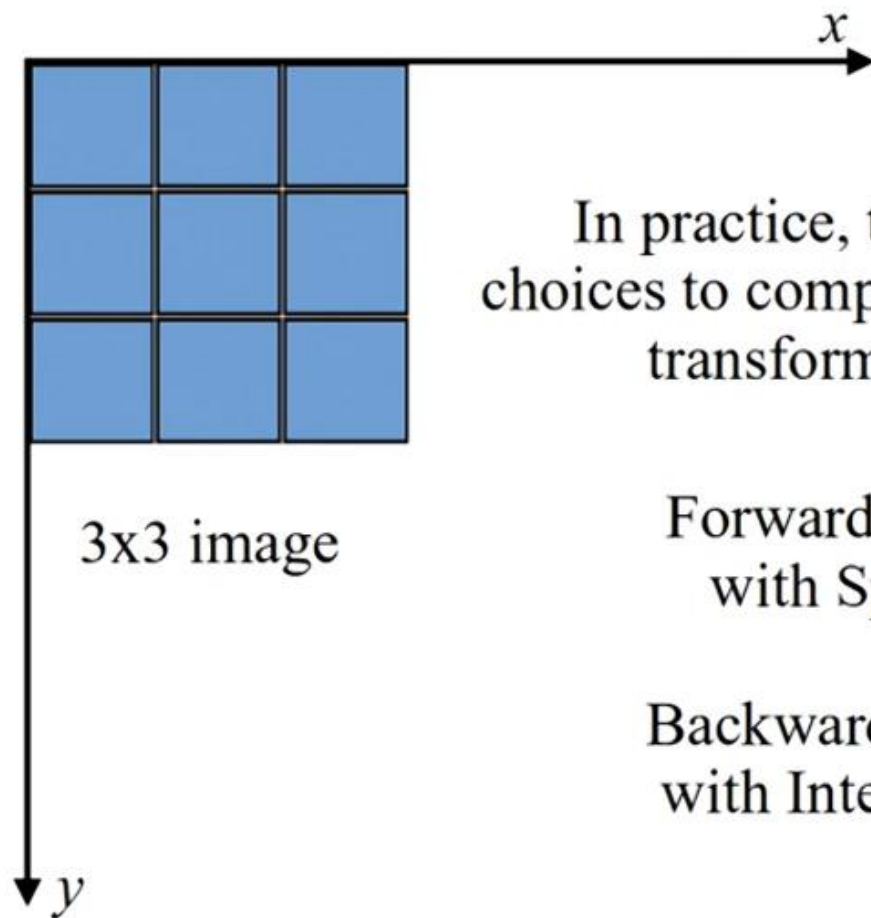
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



75% of the pixels remain empty

This is the equation to scale in
both directions by 2

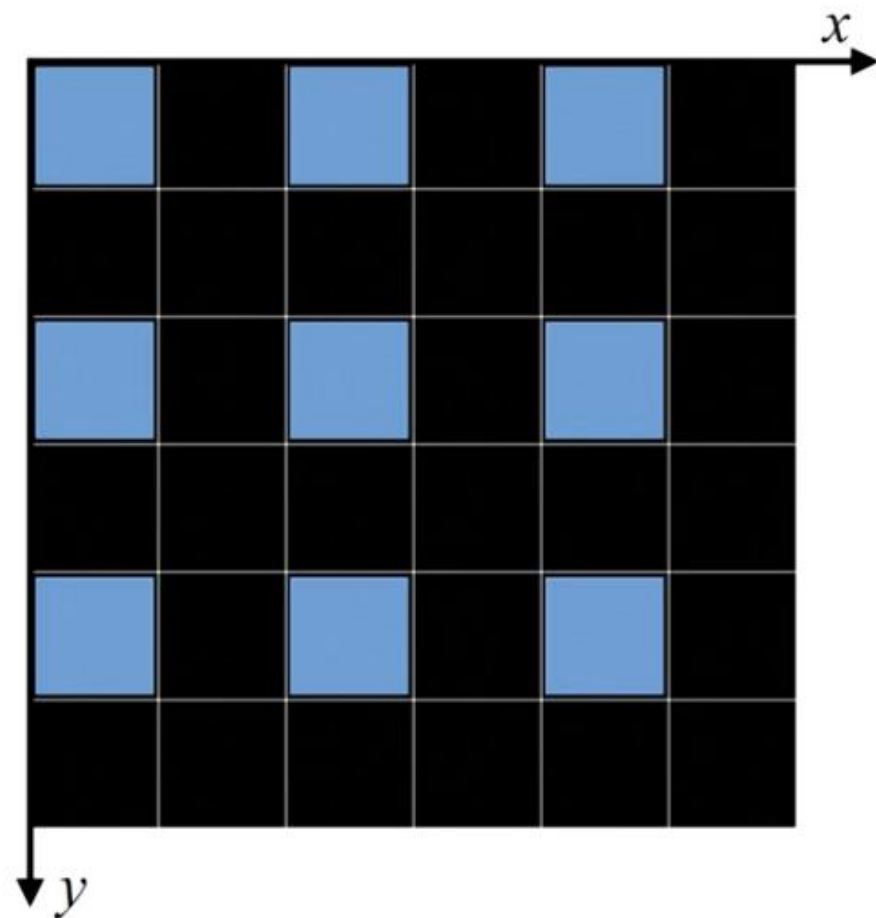
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



In practice, there are two
choices to compute the pixels of
transformed image

Forward Warping
with Splatting

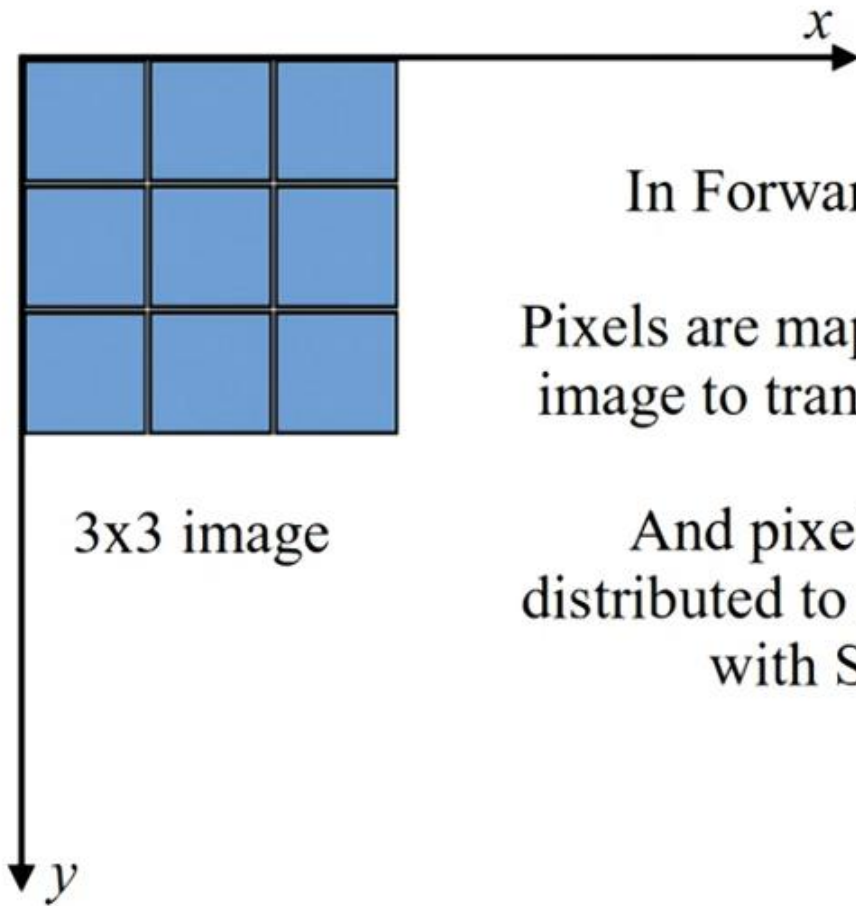
Backward Warping
with Interpolation



75% of the pixels remain empty

This is the equation to scale in
both directions by 2

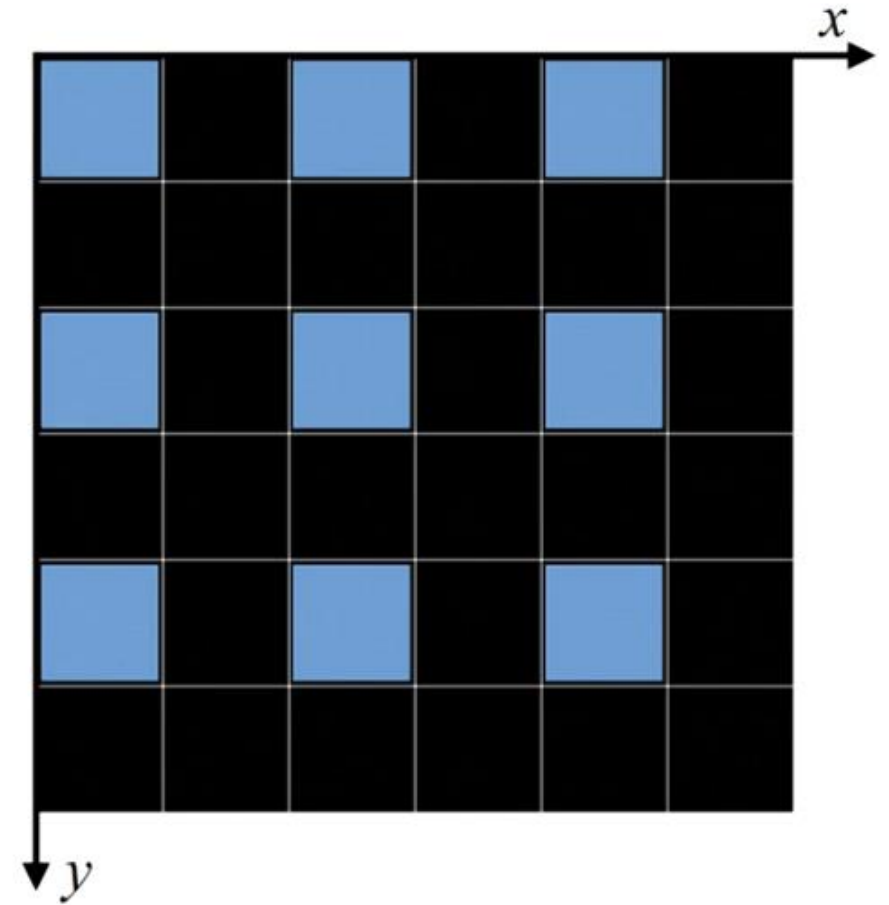
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



In Forward Warping,

Pixels are mapped from input
image to transformed image

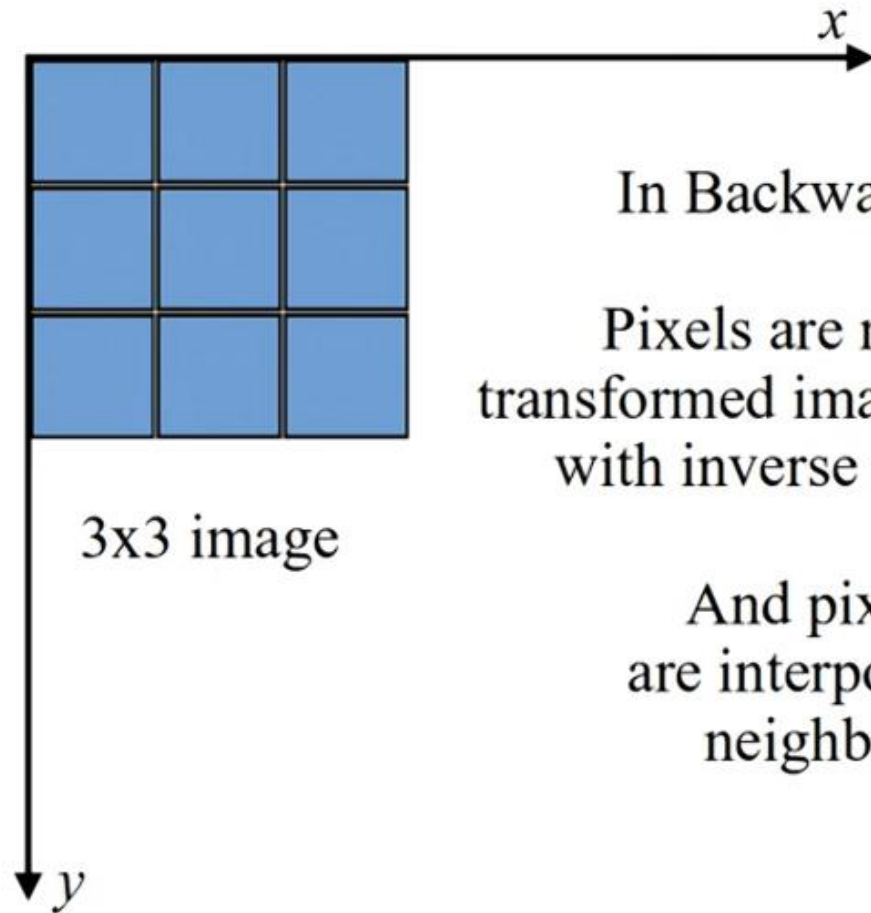
And pixel values are
distributed to neighbor pixels
with Splatting



75% of the pixels remain empty

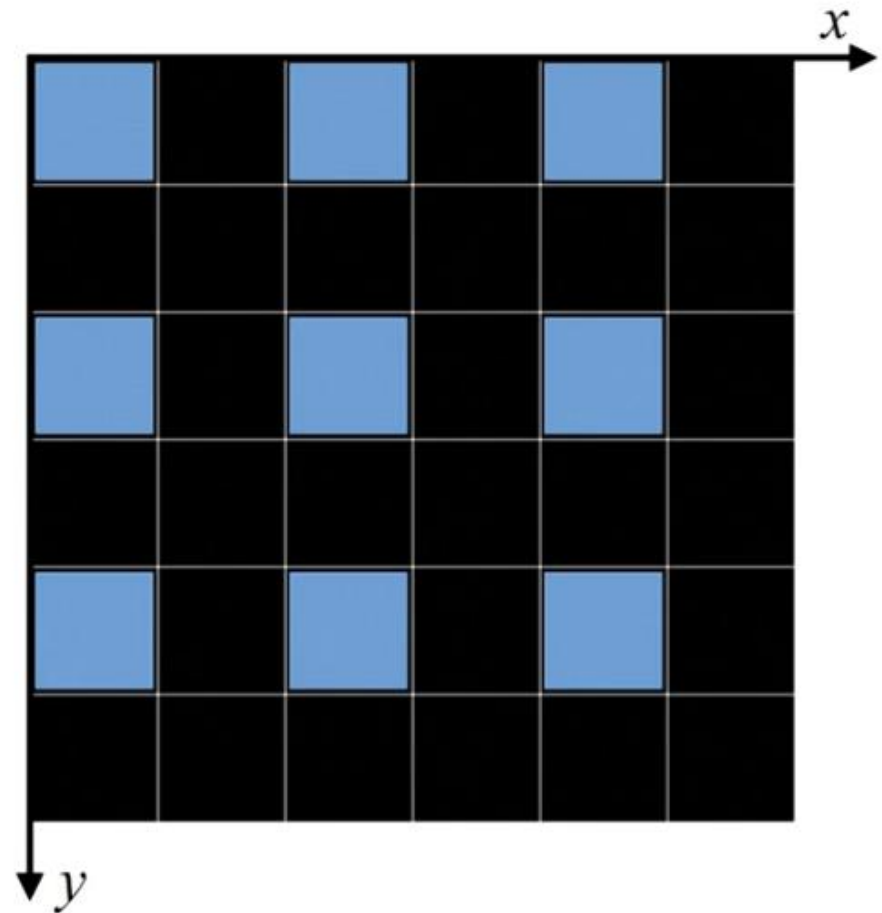
This is the equation to scale in
both directions by 2

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



In Backward Warping,
Pixels are mapped from
transformed image to input image
with inverse transformation

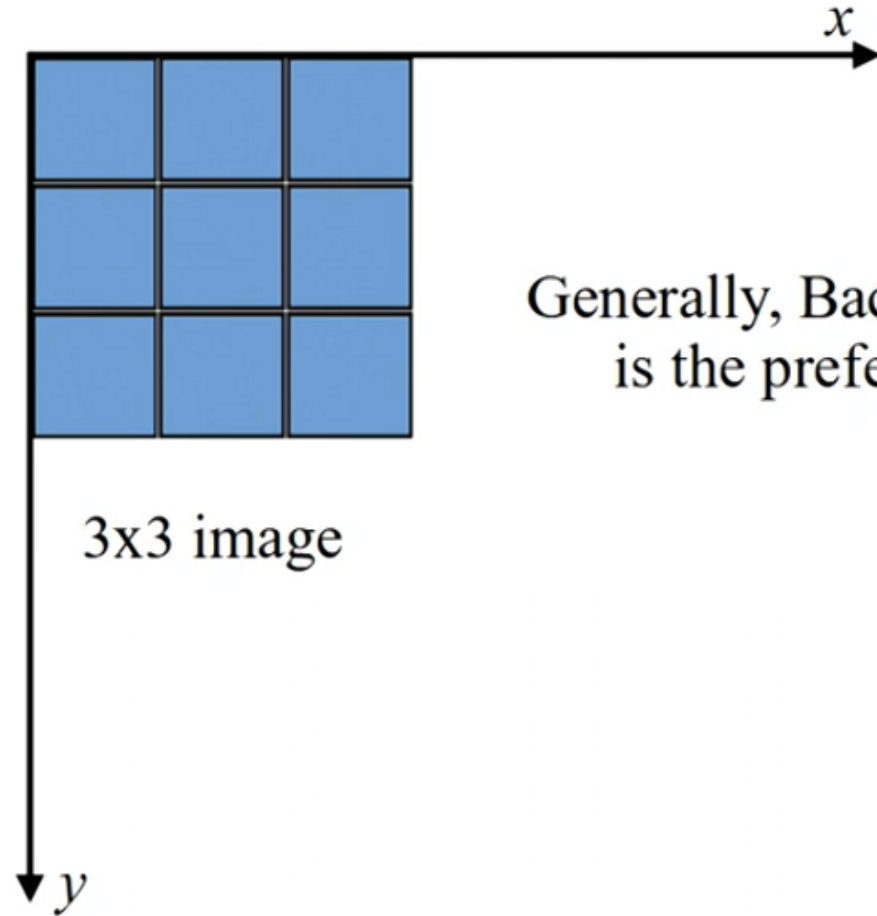
And pixel values
are interpolated using
neighbor pixels



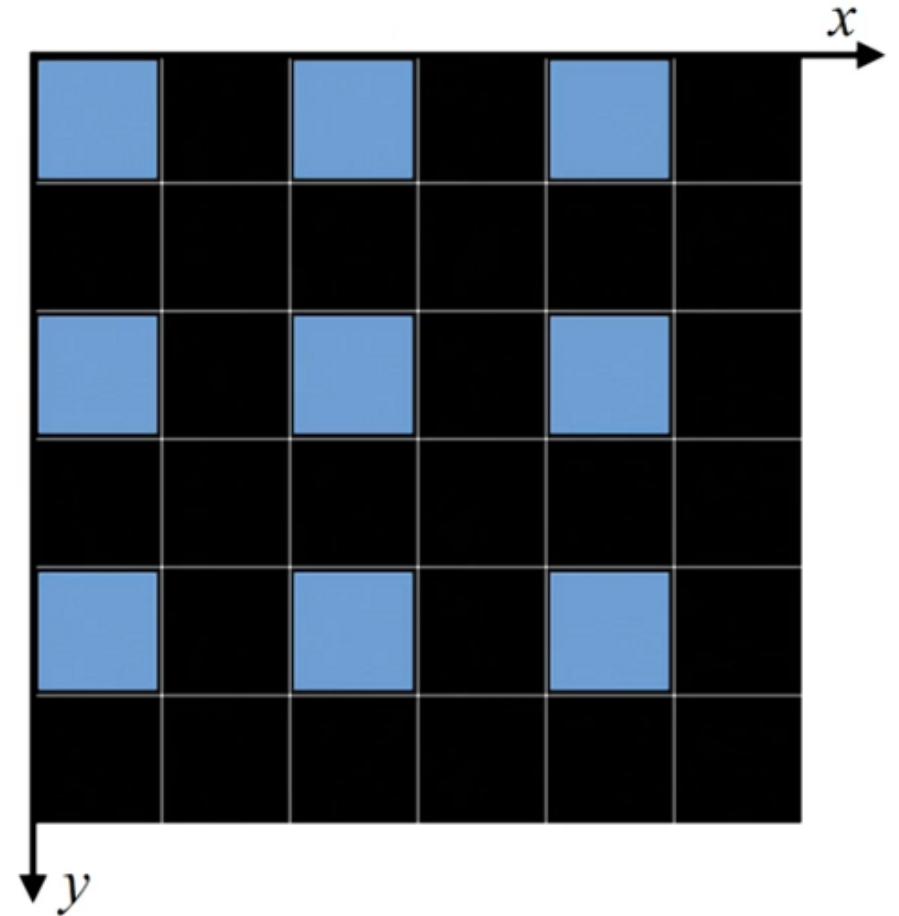
75% of the pixels remain empty

This is the equation to scale in
both directions by 2

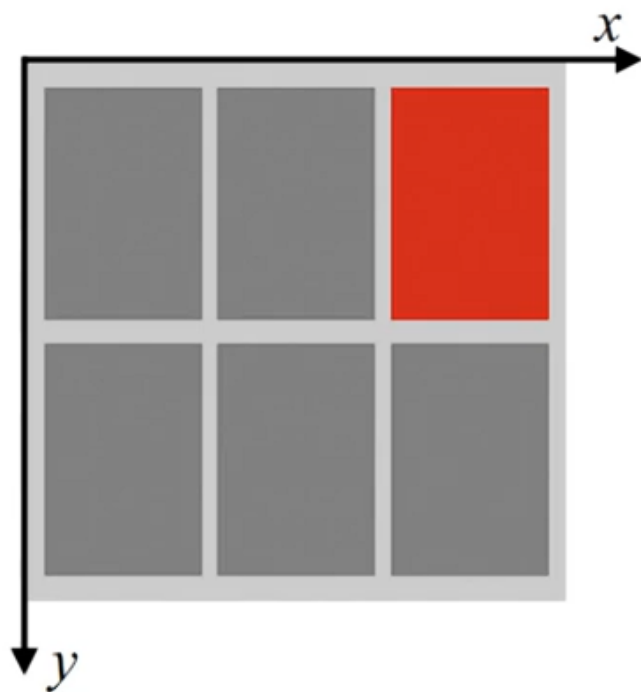
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



Generally, Backward Warping
is the preferred method

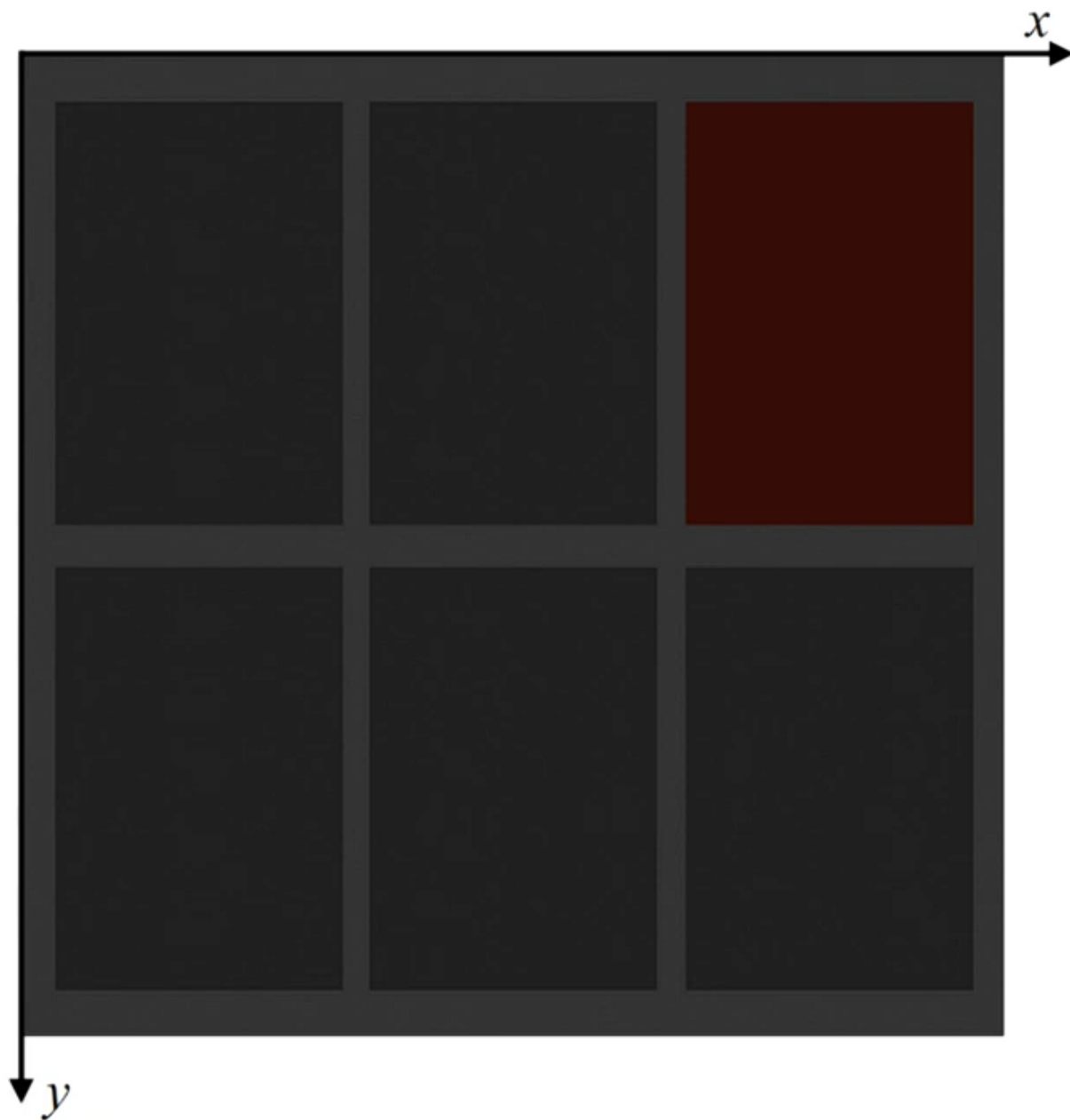


75% of the pixels remain empty



Matrix to scale image
uniformly by 2

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

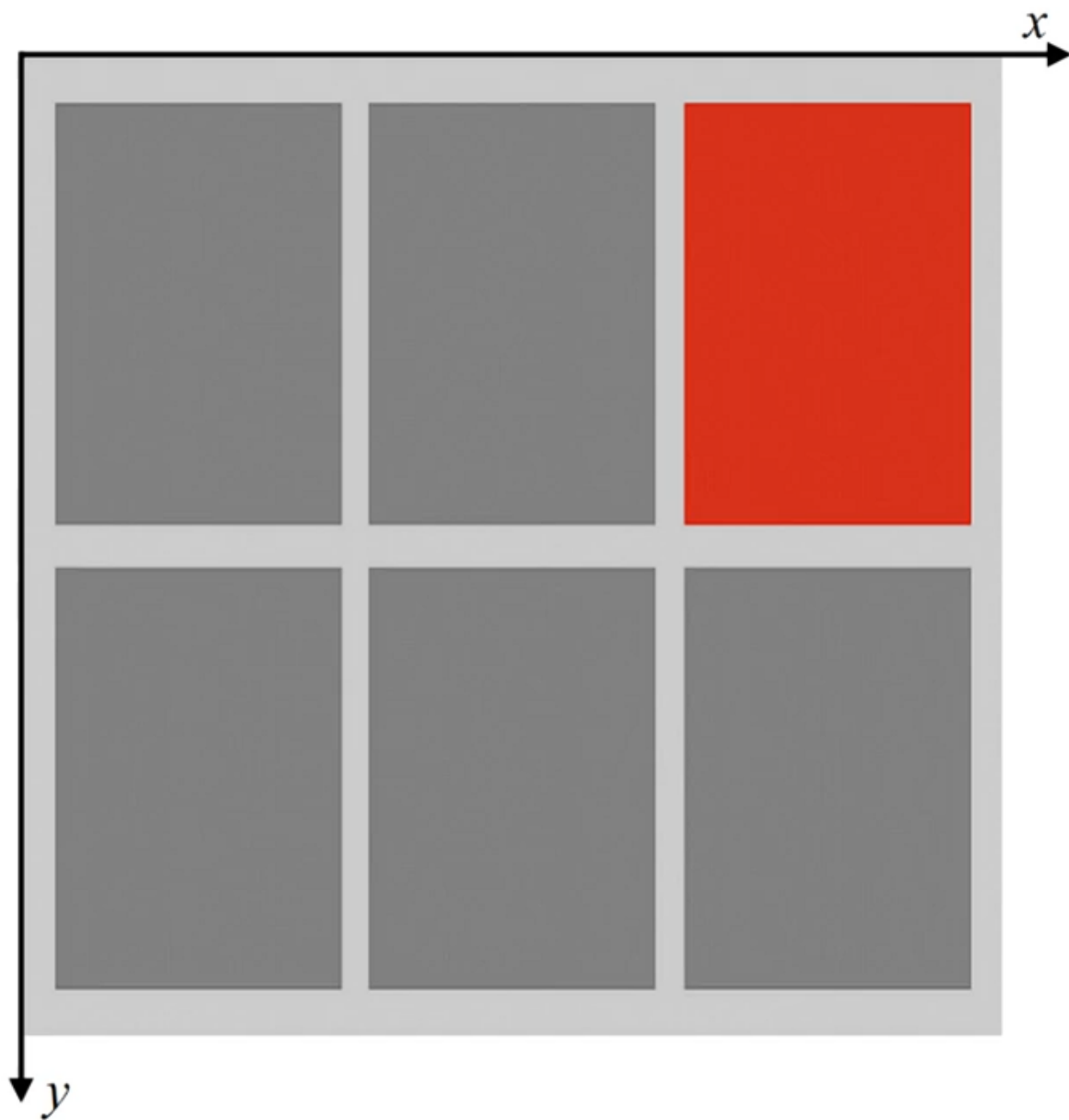


Matrix to scale image
uniformly by 2

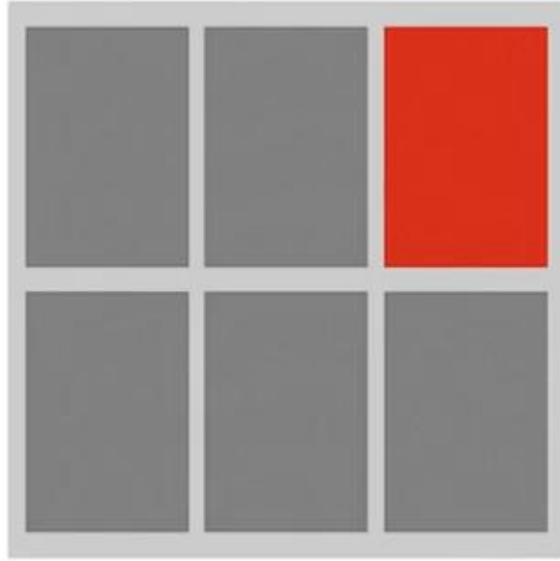
$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

This is the result of scale
transformation

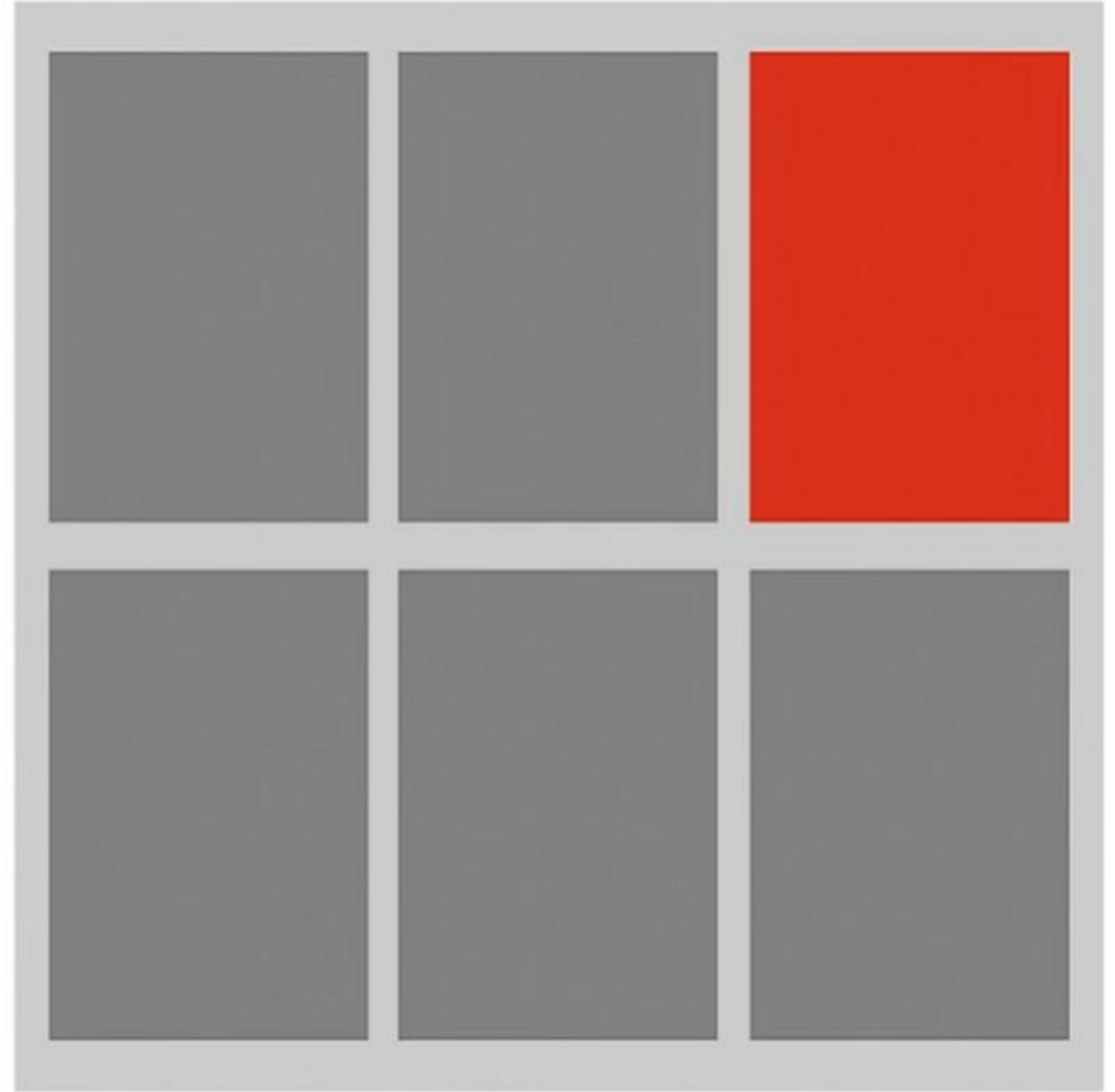
Average brightness is low due
to missing pixels



Result of Backward Warping
with interpolation



Original Image



Scaled (Resized) Image

Reference

- https://www.youtube.com/watch?v=GdL2dBucHp8&list=PLcEc12xiYZ7r24dJWP4GYwwXUgGS-wl4_&index=11